SMART IRRIGATION PREDICTIVE MAINTENANCE SYSTEM

Target Asset Specification & Sensor Placement Guide

Target Machine Model: Industrial Horizontal End-Suction Centrifugal Irrigation Pump (Grundfos NB / Kirloskar DB Series)
Standard Reference: ISO 10816-3 (Class II Medium Industrial Machinery 15 kW – 75 kW) & ANSI/HI 9.6 Centrifugal Standards

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1. Target Machine Overview & Operational Role

The targeted asset is a heavy-duty, horizontal end-suction centrifugal pump driven by an induction electric motor and connected via a flexible mechanical coupling. In agricultural irrigation systems, this pump draws water from canals or deep wells under continuous duty cycles. Because it handles variable hydraulic heads and high discharge volumes, it is subject to severe mechanical fatigue, cavitation, coupling misalignment, and bearing wear.

2. Physical Mechanical Layout & Subsystems Walkthrough

To eliminate any ambiguity in physical installation and AI root-cause attribution, the pump assembly is structured into **four sequential mechanical/electrical zones** along the drive train:

Zone / Component	Physical Description on the Machine	Mounted Sensor(s)	Direct Failure Mode Monitored
Zone 1: Electrical Drive	3-Phase AC Induction Motor (Stator windings, main electrical terminal box).	**sensor_05 (Phase A Current)** **sensor_11 (Phase B Current)**	Motor electrical overload, phase imbalance, voltage drop, and excessive winding heat.
Zone 2: Motor Drive-End (DE) Bearing	Front cast-iron motor end-shield housing supporting the drive shaft.	**sensor_00 (Radial Vibration)** **sensor_40 (Bearing Temp)**	Bearing race flaking, unbalance, looseness, lubrication starvation, and thermal seizure.
Zone 3: Mechanical Shaft & Coupling	Flexible shaft coupling (spider / elastomeric insert) joining motor to pump shaft.	**sensor_04 (Optical/Hall RPM)** **sensor_13 (Axial Vibration)**	Coupling angular/parallel misalignment, worn flexible elements, motor slip under load.
Zone 4: Hydraulic Pump Volute & Manifold	Centrifugal volute casing with enclosed impeller, mechanical seal, suction & discharge.	**sensor_28 (Flow Meter)** **sensor_48 (Discharge Pressure)**	Hydraulic cavitation, suction strainer silt blockage, loss of prime, and closed valve faults.

Engineering Principle — Spatial Isolation: Each sensor is mounted on a distinct physical subsystem. When the AI engine calculates dynamic Z-scores, a spike on **sensor_13** unequivocally isolates a coupling problem, whereas a spike on **sensor_00** isolates motor bearing degradation. This physical separation prevents ambiguous diagnostics.

3. Complete Core-7 Sensor Instrumentation Matrix

The table below defines each of the 7 sensors, its exact physical mounting location on the pump, its engineering unit, and the international maintenance standard governing its alarm limits:

Sensor ID	Hardware / Type	Physical Mounting Point	Engineering Parameter & Unit	Standard / Threshold Rule
sensor_00	ADXL345 Accelerometer	Motor Drive-End Bearing Housing	Radial Vibration Velocity (RMS mm/s or g)	*ISO 10816-3 Class II* Zone C: >4.5 mm/s (Warning) Zone D: >7.1 mm/s (Trip)
sensor_04	Tachometer / Optical	Motor Output Shaft Collar	Rotational Speed (RPM)	Rated 1450 or 2900 RPM. Slip > 6% indicates hydraulic overload.
sensor_05	Current Clamp (ACS712)	Motor Feed Terminal Line (Phase A)	Electrical Current (Amperes RMS)	Overcurrent: >115% Full Load Current (FLA) sustained for > 5 min.
sensor_11	Current Clamp (ACS712)	Motor Feed Terminal Line (Phase B)	Phase Balance Current (Amperes)	*NEMA MG 1*: Phase imbalance > 3% triggers insulation warning.
sensor_13	ADXL345 Accelerometer	Flexible Coupling Pedestal / Guard	Axial Vibration Velocity (RMS mm/s)	*ISO 10816-3*: Axial-to-radial vibration ratio > 0.6 flags misalignment.
sensor_28	Turbine Flow (YF-S201)	Discharge Flange Manifold	Discharge Flow Rate (L/min or m^3/h)	*ANSI/HI 9.6*: Flow drop < 70% BEP indicates severe cavitation/suction choke.
sensor_40	DS18B20 Temp Probe	Bearing Outer Housing Surface	Bearing Temperature (°C)	Normal: 40–60°C. Warning: >75°C. Critical: >85°C (Grease breakdown).

4. RAG Assistant & Maintenance Work Order Generation

When the ML model flags an alarm (**WARNING > 0.2713** or **CRITICAL > 0.3169**), the inference engine passes the Attributed Sensor to the RAG Assistant. The assistant queries industrial O&M; manuals and outputs a structured engineering work order containing:

• **1. Engineering Diagnosis:** Exact failure classification (e.g., Coupling Angular Misalignment, Bearing Race Spalling, Cavitation).

• **2. Urgency & Lead Time Buffer:** Action timeframe (e.g., 30–60 minutes remaining before physical functional trip).

• **3. Field Inspection Checklist:** Step-by-step procedures conforming to manufacturer recommendations (e.g., Grundfos NB Section 6).

• **4. Required Spare Parts & Tools:** Standard parts specifications (e.g., SKF 6308 deep groove ball bearing, size 110 elastomeric spider insert, dial indicator).

5. Hardware Scale Test-Bench Verification

The hardware team's benchtop prototype uses a 12V DC centrifugal pump loop with matching physical sensors. Controlled fault injection tests (such as throttling the ball valve for discharge blockage or adding an eccentric mass for coupling unbalance) produce the identical relative Z-deviation signatures as the full-scale industrial asset, proving end-to-end detection to the defense committee.